

Improving Credit Default Prediction via NLP-Based Sentiment Analysis

Course: MF810 Advanced Programming: Data Structures and Algorithms

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Date: February 18, 2026

1. Introduction & Motivation

Credit risk modeling is a fundamental component of financial stability. Financial institutions quantify Expected Loss (EL) using three key components:

- **Probability of Default (PD)**
- **Loss Given Default (LGD)**
- **Exposure at Default (EAD)**

Under regulatory frameworks such as Basel II and Basel III, traditional credit scoring models primarily rely on structured financial variables, including Debt-to-Income ratios and credit scores.

However, these models typically ignore soft information embedded in unstructured borrower-provided text, such as loan descriptions or job titles. With advances in Natural Language Processing (NLP), it is now feasible to quantify such information and test whether it improves predictive performance.

This project aims to build a hybrid credit scoring framework that integrates traditional financial indicators with NLP-derived sentiment and textual features.

2. Research Questions

Goal 1 — Foundation

Build a complete Expected Loss (EL) framework:

- **PD Model: Logistic Regression**
- **LGD Model: Beta Regression**
- **EAD Model: Linear regression using Credit Conversion Factors (CCF)**

Goal 2 — Innovation

Evaluate whether unstructured text data improves predictive performance.

Does incorporating NLP-derived features (e.g., sentiment scores and TF-IDF vectors) significantly improve the Gini coefficient, AUC, and overall discriminatory power of the PD model?

3. Data Description

We will use the Lending Club Loan Dataset (2007–2018).

3.1 Dataset Overview

- ~2,260,701 observations
- ~151 variables
- Includes both structured and unstructured features

3.2 Key Variables

Target Variable

- `loan_status` → converted into binary Default vs. Non-Default

Structured Features

- `loan_amnt`
- `int_rate`
- `annual_inc`
- `dti`
- `revol_util`

Unstructured Features

- `desc` (Borrower loan description)
- `emp_title` (Job title, high cardinality)
- `title` (Loan title)

	desc	emp_title	title
0	Borrower added on 12/22/11 > I need to upgrade my business technologies. 	NaN	Computer
1	Borrower added on 12/22/11 > I plan to use this money to finance the motorcycle i am looking at. I plan to have it paid off as soon as possible/when i sell my old bike. I only need this money because the deal im looking at is to good to pass up. Borrower added on 12/22/11 > I plan to use this money to finance the motorcycle i am looking at. I plan to have it paid off as soon as possible/when i sell my old bike.I only need this money because the deal im looking at is to good to pass up. I have finished college with an associates degree in business and its takingmeplaces 	Ryder	bike
2		NaN	real estate business
3	Borrower added on 12/21/11 > to pay for property tax (borrow from friend, need to pay back) & central A/C need to be replace. I'm very sorry to let my loan expired last time. 	AIR RESOURCES BOARD	personel
4	Borrower added on 12/21/11 > I plan on combining three large interest bills together and freeing up some extra each month to pay toward other bills. I've always been a good payor but have found myself needing to make adjustments to my budget due to a medical scare. My job is very stable, I love it. 	University Medical Group	Personal
...	...	...	...
466280		NaN	Financial Advisor
466281		NaN	Chief of Interpretation (Park Ranger)
466282	Borrower added on 12/06/13 > I am going to consolidate a credit card debt from medical and moving expenses with a loan I already have. I have paid everything on time for years, I just need to eliminate this high interest credit card. 	patrol	Debt consolidation
466283		NaN	Server Engineer Lead
466284	Borrower added on 12/04/13 > I will like a loan to fix our cars and paid our house property tax 	NaN	Other

3.3 Data Challenges

- High-cardinality categorical variables
- Sparse and noisy text fields
- Non-random missing patterns (e.g., `mths_since_last_major_derog`)
- Large-scale data processing requirements

4. Methodology

The project will follow a structured data science workflow and will be implemented in a Dockerized Jupyter environment for reproducibility.

4.1 Data Storage & Processing

- Data stored in Parquet (columnar format)
- Processed using Apache Spark via PySpark
- Parallelized processing using MapReduce paradigms

4.2 Baseline Model: Structured Features Only

Step 1: Feature Engineering

- Missing value imputation
- Categorical encoding via:

- Weight of Evidence (WoE)
- Information Value (IV) selection

Step 2: Modeling

- PD: Logistic Regression
- LGD: Beta Regression (bounded between 0 and 1)
- EAD: Linear regression for Credit Conversion Factor

This baseline model serves as the benchmark.

4.3 Advanced Hybrid Model: Structured + NLP Features

4.3.1 Text Cleaning & Normalization

Parallel text preprocessing including:

- HTML tag removal
- Lowercasing
- Stopword removal
- Punctuation cleaning

High-cardinality job titles (`emp_title`) will be clustered into occupational groups before WoE transformation.

4.3.2 NLP Feature Extraction

1. Sentiment Analysis

- Apply sentiment scoring to `desc`
- Output numerical score $\in [-1, 1]$
- Interpreted as a borrower psychological proxy

2. TF-IDF Vectorization

- Applied to `desc` and `title`
- Converts text into sparse numerical vectors
- Captures topical risk signals (e.g., “medical bills”, “debt consolidation”)

4.3.3 PD Model Enhancement

NLP-derived features will be incorporated into the baseline logistic regression PD model to construct an advanced hybrid model.

We will compare:

- **Baseline PD Model (Structured Features Only)**
- **Hybrid PD Model (Structured + NLP Features)**

The primary comparison will focus on model discrimination performance, evaluated using:

- **Area Under the ROC Curve (AUC)**
- **Gini Coefficient**
- **Kolmogorov–Smirnov (KS) Statistic**
- **Lift Analysis**

Beyond statistical discrimination, we will also assess the downstream economic implications by evaluating the implied Expected Loss (EL):

$$EL = PD * LGD * EAD$$

5. Model Evaluation

Performance will be evaluated using:

- **ROC Curve**
- **AUC**
- **Gini Coefficient**
- **Kolmogorov-Smirnov (KS) Statistic**
- **Lift analysis**

The key metric of innovation will be:

Incremental improvement in AUC and Gini after adding soft information features.

6. Technical Implementation

Component	Technology
Storage	Parquet
Big Data Processing	Apache Spark
NLP	TF-IDF, Sentiment Analysis
Modeling	Logistic Regression, Beta Regression

7. Expected Contributions

This project contributes in three dimensions:

1. Methodological Contribution
Demonstrates integration of econometric credit models with NLP techniques.
2. Practical Contribution
Provides a scalable, reproducible hybrid credit scoring pipeline.
3. Empirical Contribution
Quantifies the economic value of soft information in improving PD discrimination power.

8. Conclusion

This project bridges traditional regulatory credit modeling and modern NLP-driven data science. By systematically comparing structured-only models with hybrid models, we aim to empirically validate whether borrower-generated text contains incremental predictive value in credit risk modeling.